

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(\cos x)}{e^x+1} dx$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(\cos x)}{e^x+1} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(\cos -x)}{e^{-x}+1} dx$$

$$2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\ln(\cos x)}{e^x+1} + \frac{\ln(\cos x)}{e^{-x}+1} \right) dx$$

$$2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(\cos x) + e^x \ln(\cos x)}{e^x+1} dx$$

$$2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \ln(\cos x) dx$$

$$4 \int_0^{\frac{\pi}{2}} \ln(\cos x) dx$$

$$4 \int_0^{\frac{\pi}{2}} \ln(\cos(\frac{\pi}{2} - x)) dx$$

$$2 \int_0^{\frac{\pi}{2}} (\ln(\cos(\frac{\pi}{2} - x)) + \ln(\cos x)) dx$$

$$2 \int_0^{\frac{\pi}{2}} \ln(\sin x \cos x) dx$$

$$2 \int_0^{\frac{\pi}{2}} \ln\left(\frac{1}{2} \sin 2x\right) dx$$

$$2 \int_0^{\frac{\pi}{2}} \left(\ln \frac{1}{2} + \ln(\sin 2x) \right) dx$$

$$u := 2x$$

$$du = 2dx$$

$$2x \ln \frac{1}{2} \Big|_0^{\frac{\pi}{2}} + \int_0^{\pi} \ln(\sin u) du$$